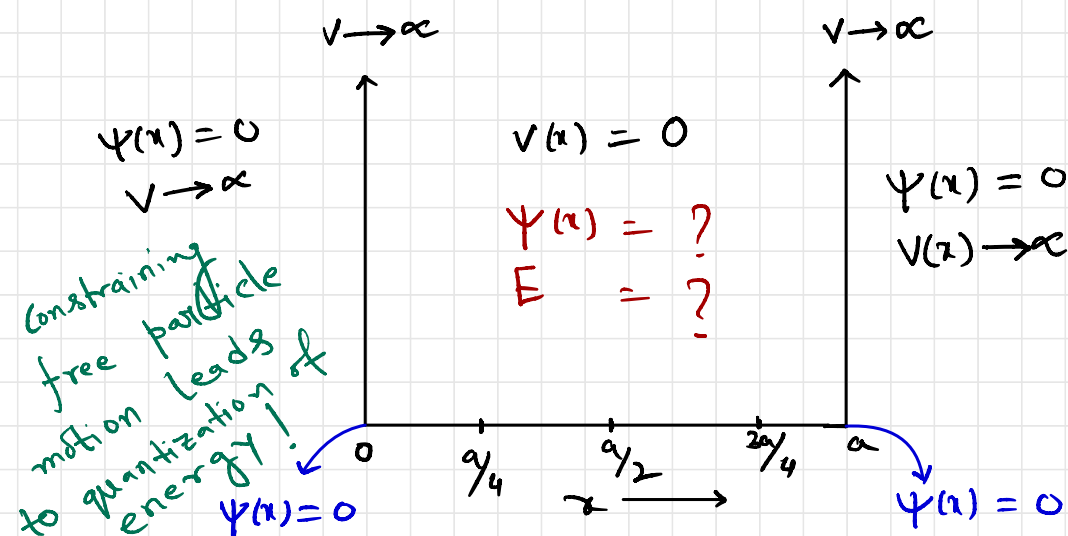


PARTICLE IN ONE-DIMENSIONAL BOX

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \underbrace{V(x)}_0 \right] \Psi(x) = E \Psi(x)$$

Wavefunction

$$\Psi(x) = C \cos kx + D \sin kx$$

By applying Boundary Conditions

$$\left. \begin{array}{l} \Psi(x) = 0 \quad \text{at } x = 0 \\ \Psi(x) = 0 \quad \text{at } x = a \end{array} \right\}$$

Boundary Conditions OR BC

$$\left| \begin{array}{l} k^2 = \frac{2mE}{\hbar^2} \\ \Rightarrow E = \frac{k^2 \hbar^2}{2m} \end{array} \right.$$

Derive $\Psi(x)$ inside the 1D box

Applying the 1st BC

$$\Psi(x) = D \sin kx$$

At $x=0$, $\Psi(x)=0$

$$\Psi(x) = \underbrace{C \cos kx + D \sin kx}_{0 \text{ as } \sin 0 = 0} = 0$$

ONLY
if $C=0$ as $\cos 0 = 1$

Applying the 2nd BC

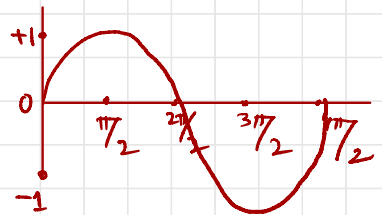
$$\underline{\underline{\Psi(x)=0 \text{ at } x=a}}$$

$$\Psi(x=a) = \cancel{D} \sin ka = 0 = \cancel{D} \sin n\pi$$

$$\Rightarrow ka = n\pi$$

For
 $n = 1, 2, 3, \dots$

$$\Rightarrow k = \frac{n\pi}{a} = \frac{n\pi}{a}$$



$$\left| \begin{aligned} k^2 &= \frac{2mE}{\hbar^2} \\ \Rightarrow E &= \frac{k^2 \hbar^2}{2m} = \frac{(n\pi)^2 \hbar^2}{a^2 \times 2m} = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \left(\frac{\hbar^2}{2\pi^2} \right) \\ &= \frac{n^2 \hbar^2}{8ma^2} \end{aligned} \right.$$

$$E = \frac{n^2 \hbar^2}{8ma^2}$$

$n: 1, 2, 3, \dots$
 $\uparrow$
quantum numbers
of various quantum states

ENERGY

Energy is quantized!

$$\psi(x) = D \sin kx$$

$$k = \frac{n\pi}{a}$$

$$\Rightarrow \psi(x) = D \sin \frac{n\pi}{a} x$$



?

D: Arbitrary
Constant

By making Probability = 1

Probability

$$P(x) = \int_0^a \psi^*(x) \psi(x) dx = 1$$

Normalization
Condition

$$\int_0^a P(x) dx = \int_0^a D \sin \frac{n\pi}{a} x \times D \sin \frac{n\pi}{a} x dx$$

$$\int_0^a \psi^*(x) \psi(x) dx = 1$$

$$\Rightarrow \int_0^a D^2 \sin^2 \left(\frac{n\pi}{a} x \right) dx = 1$$

$$\Rightarrow \int_0^{n\pi} \sin^2 z \times \frac{a}{n\pi} dz = 1$$

Consider

$$\frac{n\pi}{a} x = z$$

$$\Rightarrow \frac{n\pi}{a} dx = dz$$

$$\Rightarrow dx = \frac{a}{n\pi} dz$$

Limits

OLD	$x = x$	$z = 0$	$x = a$
NEW	$z = \frac{n\pi x}{a}$	$z = 0$	$z = n\pi$

$$\Rightarrow \frac{a}{n\pi} \int_0^{n\pi} \sin^2 z dz = 1$$

$$\Rightarrow \frac{a}{n\pi} \int_0^{n\pi} \frac{1}{2} (1 - \cos 2z) dz = 1$$

$\sin^2 z = \frac{1}{2}(1 - \cos 2z)$

$$\Rightarrow \frac{a}{n\pi} \left\{ \left. \frac{z}{2} \right|_0^{n\pi} - \frac{1}{2} \int_0^{n\pi} \cos 2z dz \right\} = 1$$

$$\Rightarrow \frac{a}{n\pi} \frac{1}{2} [n\pi - 0] = 1 \Rightarrow D = \sqrt{\frac{2}{a}}$$

$n = 1, 2, 3, \dots$

$a =$ length
of the
box

$m =$ mass
of the
particle

$h =$ Planck's Constant

$$\Psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

$$E = \frac{n^2 h^2}{8ma^2}$$